

$$1. \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ & a_{22} & \cdots & a_{2n} \\ & & \ddots & \vdots \\ & & & a_{nn} \end{vmatrix} =$$

$$7. \begin{vmatrix} a_{11} & & & \\ a_{21} & a_{22} & & \\ \vdots & \vdots & \ddots & \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix} =$$

$$13. \begin{vmatrix} a_{11} & 0 & \cdots & 0 \\ 0 & a_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_{nn} \end{vmatrix} =$$

$$2. \begin{vmatrix} a_{11} & \cdots & a_{1,n-1} & a_{1n} \\ a_{21} & \cdots & a_{2,n-1} & 0 \\ \vdots & & \vdots & \vdots \\ a_{n1} & 0 & \cdots & 0 \end{vmatrix} =$$

$$8. \begin{vmatrix} 0 & \cdots & 0 & a_{1n} \\ 0 & \cdots & a_{2,n-1} & a_{2n} \\ \vdots & & \vdots & \vdots \\ a_{n1} & \cdots & a_{n,n-1} & a_{nn} \end{vmatrix} =$$

$$14. \begin{vmatrix} 0 & \cdots & 0 & a_{1n} \\ 0 & \cdots & a_{2,n-1} & 0 \\ \vdots & & \vdots & \vdots \\ a_{n1} & \cdots & 0 & 0 \end{vmatrix} =$$

$$3. \begin{vmatrix} \mathbf{A} & \mathbf{O} \\ \mathbf{O} & \mathbf{B} \end{vmatrix} =$$

$$9. \begin{vmatrix} \mathbf{A} & * \\ \mathbf{O} & \mathbf{B} \end{vmatrix} =$$

$$15. \begin{vmatrix} \mathbf{A} & \mathbf{O} \\ * & \mathbf{B} \end{vmatrix} =$$

$$4. \begin{vmatrix} \mathbf{O} & \mathbf{A} \\ \mathbf{B} & \mathbf{O} \end{vmatrix} =$$

$$10. \begin{vmatrix} * & \mathbf{A} \\ \mathbf{B} & \mathbf{O} \end{vmatrix} =$$

$$16. \begin{vmatrix} \mathbf{O} & \mathbf{A} \\ \mathbf{B} & * \end{vmatrix} =$$

$$5. \begin{bmatrix} a_1 & & \\ & a_2 & \\ & & a_3 \end{bmatrix}^n =$$

$$11. \begin{bmatrix} a_1 & & \\ & a_2 & \\ & & a_3 \end{bmatrix}^{-1} =$$

$$17. \begin{bmatrix} & a \\ & b \\ c & \end{bmatrix}^{-1} =$$

$$6. \begin{bmatrix} O & B \\ C & O \end{bmatrix}^{-1} =$$

$$12. \begin{bmatrix} B & O \\ O & C \end{bmatrix}^n =$$

$$18. \begin{bmatrix} B & O \\ O & C \end{bmatrix}^{-1} =$$

	伴随	逆	转置	行列式
伴随	$(A^*)^* =$	$(A^*)^{-1} =$	$(A^*)^T =$	$ A^* =$
逆	$(A^{-1})^* =$	$(A^{-1})^{-1} =$	$(A^{-1})^T =$	$ A^{-1} =$
转置	$(A^T)^* =$	$(A^T)^{-1} =$	$(A^T)^T =$	$ A^T =$
行列式		$ A ^{-1} =$	$ A ^T =$	$ A =$
数	$(kA)^* =$	$(kA)^{-1} =$	$(kA)^T =$	$ kA =$
穿脱原则	$(AB)^* =$	$(AB)^{-1} =$	$(AB)^T =$	$ AB =$
加法			$(A+B)^T =$	
交换	$AA^* =$	$AA^{-1} =$		

$$1. [E_i(k)]^{-1} =$$

$$3. [E_{ij}(k)]^{-1} =$$

$$5. E_{ij}^{-1} =$$

$$2. r(A^*) =$$

$$4. |A| \text{ 逆序数:}$$

$$6. E + A^3 =$$

$$3. E - A^3 =$$

$$5. AB - 2B - 4A = 0 \Leftrightarrow$$

$$7. \begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} =$$

$$6. A^{-1}/A \text{ 与 } A^* \text{ 关系:}$$

$$7. \text{施密特正交化:}$$

设 A 是 $m \times n$ 矩阵, B 满足有关矩阵运算要求的矩阵, 则

1. $r(A)$ 范围

2. $r(A)$ 与哪些矩阵的秩相等

3. $r(AB)$ 范围

4. $r([A, B])$ 范围

5. 若 C 是 $s \times t$ 矩阵, 则

6. 若 $A \sim B$, 则

7. 若 $A_{m \times n} B_{n \times s} = \mathbf{O}$, 则
 - $r(A) + r(B)$
 - 若 $\mathbf{A}$ 和 $\mathbf{B}$ 为方阵, 则
 - 且 $A B$ 非零, 则
8. 齐次方程组有解条件

9. 同解方程组

10. 公共解

11. 矩阵等价

12. 正交矩阵

13. 非齐次方程组有解条件

1.
$$\begin{vmatrix} a_1 & b_1 & & & \\ c_2 & a_2 & b_2 & & \\ & c_3 & a_3 & b_3 & \\ & & \ddots & \ddots & \ddots \\ & & & c_{n-1} & a_{n-1} & b_{n-1} \\ & & & & c_n & a_n \end{vmatrix} =$$